

ENGINEERING PHYSICS

COMPLETE EXAM REVISION HANDBOOK

Quantum Theory • Atomic Structure • Quantum Mechanics
Applications • Quantum Computing

All Derivations • All Formulas • PYQ-Style Questions • Professor Traps • Memory Tricks

UNIT I — ORIGIN OF QUANTUM THEORY

1. Blackbody Radiation

Definition

A blackbody is an ideal object that absorbs all incident electromagnetic radiation and emits radiation solely due to its temperature. The radiation emitted is called blackbody (or cavity) radiation.

Physical Meaning (Intuitive)

Imagine a furnace with a tiny hole. Any radiation entering the hole gets absorbed inside (hence 'black'). The radiation leaking out depends purely on temperature—not on the material. A perfect blackbody is the most efficient radiator possible at any given temperature.

Key Observations

- As temperature increases, peak wavelength shifts to shorter values (blue shift).
- Total power radiated increases rapidly with temperature.
- Classical physics FAILED to explain the spectrum—gave the 'ultraviolet catastrophe.'

Laws of Blackbody Radiation

Stefan-Boltzmann Law

$$E = \sigma T^4 \quad [\text{Total energy radiated per unit area per unit time}]$$

where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ (Stefan-Boltzmann constant)

Wien's Displacement Law

$$\lambda_{\text{max}} \times T = b = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$$

As temperature rises, the peak wavelength decreases inversely.

Rayleigh-Jeans Law (Classical — FAILS at short λ)

$$u(\lambda) = 8\pi kT / \lambda^4 \quad [\text{Ultraviolet Catastrophe at small } \lambda]$$

Planck's Radiation Law (Quantum — Correct)

$$u(\nu) = (8\pi h \nu^3 / c^3) \times 1 / (e^{(h\nu/kT)} - 1)$$

$$u(\lambda) = (8\pi hc / \lambda^5) \times 1 / (e^{(hc/\lambda kT)} - 1)$$

Planck's constants: $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$, $k = 1.38 \times 10^{-23} \text{ J/K}$, $c = 3 \times 10^8 \text{ m/s}$

Planck's Quantum Hypothesis

Planck proposed that energy exchange between radiation and matter occurs in discrete packets called QUANTA (photons). Each quantum has energy:

$$E = h\nu = hc/\lambda$$

Planck's hypothesis was revolutionary: energy is NOT continuous—it comes in integer multiples of $h\nu$. This broke classical physics and launched quantum theory.

Derivation of Planck's Law (Outline)

1. Assume oscillators in the cavity wall have energies that are multiples of $h\nu$: $E_n = nh\nu$ ($n = 0, 1, 2, \dots$)
2. Use Boltzmann distribution to find average energy: $\langle E \rangle = h\nu / (e^{(h\nu/kT)} - 1)$
3. Multiply by density of modes ($8\pi\nu^2/c^3$) to get energy density: $u(\nu) = (8\pi h\nu^3/c^3) / (e^{(h\nu/kT)} - 1)$

Limiting Cases (Show mastery!)

- At high T or low ν : $e^{(h\nu/kT)} \approx 1 + h\nu/kT$, so Planck $\rightarrow$ Rayleigh-Jeans $\checkmark$
- At low T or high ν : $e^{(h\nu/kT)} \gg 1$, so Planck $\rightarrow$ Wien's Law $\checkmark$

⚠ TRAP: Many students confuse 'blackbody' with a black-colored object. A blackbody is defined by its absorption property, not color. The Sun approximates a blackbody.

✓ TRICK: Memory: Stefan-Boltzmann $\rightarrow$ 'E equals sigma T to the 4th.' Wien $\rightarrow$ ' $\lambda_{\text{max}} T = b$ (b for 2.898).'

Question Type	Likely Questions
2-mark	Define blackbody. State Planck's hypothesis. State Wien's law.
5-mark	Explain ultraviolet catastrophe and how Planck resolved it.
10-mark	Derive Planck's radiation formula. Show limiting cases.
Numerical	Find λ_{max} for a body at 6000 K. Find total intensity at 3000 K.

2. Photoelectric Effect

Definition

When electromagnetic radiation of sufficient frequency falls on a metal surface, electrons are emitted from the surface. This phenomenon is called the photoelectric effect.

Experimental Observations

- Emission of photoelectrons occurs only if the incident frequency $\nu \geq$ threshold frequency ν_0 .
- Number of photoelectrons is proportional to the intensity of incident light.
- Maximum kinetic energy of photoelectrons depends ONLY on frequency, NOT on intensity.
- Emission is instantaneous (no time delay).

Classical Failure

Classical wave theory predicted: (1) any frequency should work given enough time, (2) KE should depend on intensity, (3) there should be a time delay. All three predictions are WRONG.

Einstein's Photoelectric Equation

$$KE_{\text{max}} = h\nu - \phi = h\nu - h\nu_0$$

$$eV_s = h\nu - \phi \quad [\text{where } V_s = \text{stopping potential}]$$

$$\phi = h\nu_0 \quad [\text{Work function} = \text{minimum energy to release electron}]$$

Key Quantities

Quantity	Symbol	Definition	Unit
Work function	ϕ or W	Minimum energy to eject electron	eV or J
Threshold frequency	ν_0	Minimum frequency for emission	Hz
Threshold wavelength	λ_0	Maximum wavelength for emission	m
Stopping potential	V_s	Potential to stop fastest electrons	Volt
Max KE	$KE_{\max}$	$h\nu - \phi$	eV or J

Important Relations

$$\phi = h\nu_0 = hc/\lambda_0$$

$$\lambda_0 = hc/\phi$$

$$V_s = (h/e)(\nu - \nu_0) = h(\nu - \nu_0)/e$$

1 eV = 1.6×10^{-19} J. Always convert work function from eV to J for SI calculations.

Solved Numerical Example

A metal has work function $\phi = 2.0$ eV. Find stopping potential for light of $\lambda = 3000$ Å.

Step 1: $\nu = c/\lambda = (3 \times 10^8)/(3 \times 10^{-7}) = 10^{15}$ Hz

Step 2: $h\nu = (6.626 \times 10^{-34})(10^{15}) = 6.626 \times 10^{-19}$ J = 4.14 eV

Step 3: $KE_{\max} = 4.14 - 2.0 = 2.14$ eV

Step 4: $V_s = KE_{\max}/e = 2.14$ V

⚠ TRAP: Students often forget to convert eV to Joules in mid-calculation. Safest: work entirely in eV using $hc = 1240$ eV·nm or $hc = 12400$ eV·Å.

✓ TRICK: Shortcut: $hc = 1240$ eV·nm = 12400 eV·Å. Use this to get $h\nu$ in eV directly: $h\nu(\text{eV}) = 12400/\lambda(\text{Å})$

3. Compton Effect

Definition

When X-rays (high-frequency photons) are scattered by free or loosely bound electrons, the scattered X-rays have a longer wavelength than the incident X-rays. This shift in wavelength is the Compton effect.

Physical Meaning

A photon acts like a particle with momentum $p = h\nu/c = h/\lambda$. When it collides with an electron, it transfers some momentum and energy to the electron, thereby decreasing its own energy (increasing its wavelength). This PROVES the particle nature of light.

Compton Shift Formula

$$\Delta\lambda = \lambda' - \lambda = (h/m_e c)(1 - \cos \theta)$$

$$\Delta\lambda = \lambda_C (1 - \cos \theta)$$

where $\lambda_C = h/(m_e c) = 2.426 \times 10^{-12} \text{ m} = 0.02426 \text{ \AA}$ [Compton wavelength]

Derivation of Compton Shift (Full)

Setup: Photon (energy $h\nu$, momentum h/λ) strikes electron at rest (mass m_e).

After collision: Photon scattered at angle θ (wavelength λ'), electron recoils at angle ϕ .

Step 1 — Conservation of Energy

$$h\nu + m_e c^2 = h\nu' + \gamma m_e c^2$$

Step 2 — Conservation of x-Momentum

$$h/\lambda = h/\lambda' \times \cos \theta + \gamma m_e v \cos \phi$$

Step 3 — Conservation of y-Momentum

$$0 = h/\lambda' \times \sin \theta - \gamma m_e v \sin \phi$$

Step 4 — Eliminate electron terms

Square and add momentum equations to eliminate ϕ . Use energy equation to eliminate γ . After algebraic manipulation:

$$\Delta\lambda = \lambda' - \lambda = (h/m_e c)(1 - \cos \theta)$$

Special Cases

Scattering Angle θ	$\cos \theta$	$\Delta\lambda$
0° (forward)	1	0 (no shift)
90° (side)	0	$\lambda_C = 0.02426 \text{ \AA}$
180° (back)	-1	$2\lambda_C = 0.04852 \text{ \AA}$ (maximum)

⚠ TRAP: $\Delta\lambda$ is INDEPENDENT of the incident wavelength—it depends only on the scattering angle. Students often add wavelength-dependence incorrectly.

✓ TRICK: Maximum Compton shift ($\theta=180^\circ$) = $2\lambda_C = 2 \times 0.02426 = 0.04852 \text{ \AA}$. This is the ONLY special value to memorize.

Unmodified Line

Photons scattered by tightly bound electrons (whole atom recoils). Since atom mass $\gg$ electron mass, $\Delta\lambda \approx 0$. This explains the unmodified (same wavelength) line in Compton spectra.

4. Pair Production and Pair Annihilation

Pair Production

A high-energy photon (γ -ray) spontaneously converts into an electron-positron pair in the presence of a nucleus (for momentum conservation).

$$\gamma \rightarrow e^- + e^+$$

$$\text{Minimum photon energy: } h\nu_{\min} = 2m_e c^2 = 2 \times 0.511 \text{ MeV} = 1.022 \text{ MeV}$$

- Requires nucleus nearby for momentum conservation.
- Excess energy appears as kinetic energy of the pair.
- Demonstrates $E = mc^2$ and photon's particle nature.

Pair Annihilation

An electron and positron (antiparticle) annihilate each other, converting their mass entirely into two photons (for momentum conservation).

$$e^- + e^+ \rightarrow 2\gamma$$

$$\text{Each photon energy} = m_e c^2 = 0.511 \text{ MeV}$$

- Two photons emitted in opposite directions (180° apart) to conserve momentum.
- Single photon cannot be produced alone (would violate momentum conservation).

Pair production converts energy $\rightarrow$ matter; pair annihilation converts matter $\rightarrow$ energy. Both demonstrate mass-energy equivalence $E=mc^2$.

⚠ TRAP: Students sometimes draw only ONE photon in annihilation diagrams—always TWO photons are needed.

5. De Broglie Hypothesis and Matter Waves

Statement

Every material particle in motion has an associated wave called a matter wave (de Broglie wave). The wavelength of this wave is:

$$\lambda = h/p = h/(mv)$$

Physical Meaning

Just as light (wave) has particle properties (photon), particles (electrons, protons) have wave properties. This is wave-particle duality extended to matter. The wavelength is only significant for very small particles (atomic scale); macroscopic objects have negligibly small λ .

Useful Forms of de Broglie Wavelength

$$\lambda = h/p = h/(mv) \quad [\text{General}]$$

$$\lambda = h/\sqrt{(2mE_K)} \quad [\text{From } E_K = p^2/2m]$$

$$\lambda = h/\sqrt{(2meV)} \quad [\text{For electron accelerated through } V \text{ volts}]$$

$$\lambda = 12.26/\sqrt{V} \text{ \AA} \quad [\text{Quick formula for electrons: } V \text{ in volts}]$$

$$\lambda = h/\sqrt{(2mkT)} \quad [\text{For thermal particles, } E_K = kT]$$

For electron accelerated through V volts: $\lambda (\text{\AA}) = 12.26/\sqrt{V}$. Memorize this for quick numericals!

Comparison: Photon vs Matter Wave

Property	Photon	Matter Particle
Energy	$E = h\nu$	$E = p^2/2m$ (non-rel.)

Property	Photon	Matter Particle
Momentum	$p = h/\lambda = h\nu/c$	$p = mv$
De Broglie λ	$\lambda = c/\nu$	$\lambda = h/p$
Speed	Always c	$v < c$

6. Wave Packets and Group Velocity

Wave Packet

A matter wave is NOT a pure sine wave of definite λ , but a superposition of waves with a range of wavelengths. This localized wave group is called a wave packet.

Phase Velocity (v_p)

$$v_p = \omega/k = v\lambda = E/(\hbar k)$$

The speed at which a phase of the wave travels. For matter waves: $v_p = E/p = (p^2/2m)/p = p/2m = v/2$. PROBLEMATIC: phase velocity $\neq$ particle velocity!

Group Velocity (v_g)

$$v_g = d\omega/dk$$

The speed at which the envelope (the packet as a whole) travels. This equals the particle velocity:

$$v_g = d\omega/dk = d(E/\hbar)/d(p/\hbar) = dE/dp = d(p^2/2m)/dp = p/m = v \quad \checkmark$$

Group velocity = particle velocity. Phase velocity = $v/2$ for a non-relativistic free particle. The wave packet travels at the correct particle speed.

⚠ TRAP: Confusing phase velocity (ω/k) with group velocity ($d\omega/dk$). Only group velocity equals particle velocity.

Quantity	Formula	Physical Meaning
Phase velocity v_p	$v_p = \omega/k = E/p$	Speed of wave crests
Group velocity v_g	$v_g = d\omega/dk = dE/dp$	Speed of wave packet = particle velocity
Relation (free particle)	$v_g = 2v_p$	Group velocity is twice phase velocity

7. Davisson-Germer Experiment

Purpose

Experimental proof of the wave nature of electrons (matter waves). Demonstrated electron diffraction, confirming de Broglie's hypothesis.

Setup

Electrons from an electron gun were accelerated through a potential V and directed at a single crystal of nickel. The scattered electrons were detected as a function of angle ϕ by a detector mounted on a circular scale.

Observation

A sharp peak in electron intensity was observed at $\phi = 50^\circ$ for $V = 54$ V. This is analogous to Bragg diffraction peaks for X-rays, proving electrons undergo wave diffraction.

Verification

Bragg condition for diffraction: $2d \sin \theta = n\lambda$, where $d = 0.91$ Å for Ni (100) planes.

From experiment: $\lambda_{\text{exp}} \approx 1.65$ Å

From de Broglie: $\lambda = 12.26/\sqrt{54} = 12.26/7.35 = 1.67$ Å ← EXCELLENT AGREEMENT ✓

The agreement between de Broglie's predicted wavelength (1.67 Å) and the experimental value (1.65 Å) is the key result. Always quote this in your answer.

⚠ TRAP: Students forget to use the special formula $\lambda = 12.26/\sqrt{V}$ (in Å, V in volts) for electrons. Using $\lambda = h/\sqrt{2meV}$ is correct but slower in exams.

8. Heisenberg Uncertainty Principle

Statement

It is fundamentally impossible to simultaneously measure both the position and momentum of a particle with arbitrary precision. The product of the uncertainties is always at least $\hbar/2$:

$$\Delta x \cdot \Delta p_x \geq \hbar/2$$

For most exam purposes, the rougher form is used:

$$\Delta x \cdot \Delta p \geq \hbar \quad (\text{or } h/2\pi) \quad [\text{Exam-standard form}]$$

Energy-Time Uncertainty

$$\Delta E \cdot \Delta t \geq \hbar/2 \approx \hbar$$

Angular Uncertainty

$$\Delta L \cdot \Delta \theta \geq \hbar$$

Physical Meaning

This is NOT a measurement error or instrument limitation. It is a fundamental property of wave-like quantum objects. The uncertainty arises because position and momentum are conjugate variables related through the wave's Fourier transform: a narrow wave packet (precise position) requires a wide range of wavelengths (uncertain momentum), and vice versa.

Derivation via Wave Packet Argument

4. A wave packet confined to Δx must contain waves of wave-numbers in range $\Delta k \approx 1/\Delta x$ (Fourier theorem).
5. Since $p = \hbar k$, we have $\Delta p = \hbar \Delta k$.
6. Therefore $\Delta x \cdot \Delta p \approx \hbar$.

Important Applications

Application	Key Formula	Result/Use
Finite nuclear size	$\Delta x \approx 10^{-15} \text{ m} \rightarrow \Delta p$	Minimum electron energy $\approx 200 \text{ MeV} \rightarrow$ no electrons in nucleus
Natural linewidth	$\Delta E \times \Delta t \approx \hbar$	Energy width of atomic transitions
Zero-point energy	$\Delta x \approx a \rightarrow \Delta p \approx \hbar/a$	$E_{\text{min}} = (\Delta p)^2/2m = \hbar^2/2ma^2$
Ground state size	Minimize $E = p^2/2m + e^2/r$	Bohr radius a_0 from HUP

Solved Example — HUP

An electron is confined in a box of width $1 \text{ \AA}$. Find its minimum kinetic energy.

$$\Delta x = 10^{-10} \text{ m} \rightarrow \Delta p \geq \hbar/\Delta x = 1.055 \times 10^{-34} / 10^{-10} = 1.055 \times 10^{-24} \text{ kg}\cdot\text{m/s}$$

$$KE_{\text{min}} = (\Delta p)^2/(2m) = (1.055 \times 10^{-24})^2/(2 \times 9.11 \times 10^{-31}) = 6.1 \times 10^{-19} \text{ J} \approx 3.8 \text{ eV}$$

⚠ TRAP: Using $\Delta x \cdot \Delta p \geq h$ (not $\hbar$) is also accepted in many textbooks. Clarify which form your professor prefers. The $\hbar$ form gives smaller KE_{min} .

Unit I — Exam Question Bank

2-Mark Questions

- Define blackbody radiation. State Planck's hypothesis.
- What is the photoelectric effect? State Einstein's equation.
- Define stopping potential and work function.
- What is Compton wavelength? Give its value.
- State de Broglie's hypothesis.
- What is a wave packet? Define group velocity.
- State Heisenberg uncertainty principle.
- What is pair production? Write the energy condition.

5-Mark Questions

- Explain the ultraviolet catastrophe and Planck's resolution.
- Describe photoelectric effect experiments and Einstein's explanation.
- Derive the Compton shift formula (outline derivation).
- Explain pair production and pair annihilation with energy conditions.
- Derive group velocity for a matter wave and show it equals particle velocity.
- Describe the Davisson-Germer experiment and its significance.

10-Mark Questions

- Derive Planck's radiation law from quantum hypothesis. Show limiting cases (Wien and Rayleigh-Jeans).
- Derive the Compton shift formula step-by-step using relativistic mechanics.
- Explain wave-particle duality. Describe de Broglie matter waves, wave packets, group and phase velocity. Give experimental evidence.

- State and explain Heisenberg uncertainty principle. Derive from wave packet analysis. Give two applications.

Common Mistakes

- ⚠ **TRAP: Unit I: Forgetting that Compton shift is independent of incident wavelength.**
- ⚠ **TRAP: Using $hc = 6.626 \times 10^{-34} \times 3 \times 10^8$ correctly: $= 1.988 \times 10^{-25} \text{ J}\cdot\text{m}$. Better shortcut: $hc = 1240 \text{ eV}\cdot\text{nm}$.**
- ⚠ **TRAP: Pair production requires a nucleus (third body) for momentum conservation—cannot happen in free space.**
- ⚠ **TRAP: Phase velocity of matter waves $= v/2$, NOT v . Group velocity $= v$. Always state which one.**

UNIT II — ATOMIC STRUCTURE

1. Thomson Atomic Model

Model Description

J.J. Thomson (1904) proposed that the atom is a uniformly distributed sphere of positive charge with electrons embedded in it like seeds in a watermelon ('plum pudding model').

- Positive charge distributed throughout the sphere of radius $\approx 10^{-10}$ m.
- Electrons embedded at equilibrium positions.
- Atom is electrically neutral overall.

Failure

Could not explain: (1) Rutherford's alpha scattering experiment showing large deflections, (2) discrete atomic spectra.

2. Rutherford's Alpha Scattering Experiment

Setup

Alpha particles (charge $+2e$, mass ≈ 4 u) from a radioactive source were directed at a thin gold foil. Scintillation detectors (ZnS screens) detected scattered alphas at various angles.

Observations

7. MOST alpha particles passed straight through (or deflected slightly): most of the atom is empty space.
8. SOME were deflected by large angles: concentrated positive charge at center.
9. A FEW (1 in 10,000) bounced almost straight back: nucleus is very small and dense.

Rutherford Nuclear Model

- Atom has a tiny, dense, positively charged nucleus (radius $\sim 10^{-15}$ m = 1 fm).
- Electrons orbit the nucleus at large distances ($\sim 10^{-10}$ m).
- Most of the atom is empty space.

Rutherford Scattering Formula

$$d\sigma/d\Omega = (Z_1 Z_2 e^2 / 4E)^2 \times 1/\sin^4(\theta/2)$$

where $d\sigma/d\Omega$ is the differential cross-section, E is kinetic energy of alpha, θ is scattering angle.

Distance of Closest Approach

$$r_{\min} = Z_1 Z_2 e^2 / (4\pi\epsilon_0 \times E_K) = Z_1 Z_2 k e^2 / E_K$$

At closest approach, all kinetic energy converts to electrostatic potential energy. This gives an upper bound on nuclear size.

Failures of Rutherford Model

- Accelerating electron should radiate and spiral into nucleus (classical EM theory predicts).
- Cannot explain discrete (line) atomic spectra.

⚠ **TRAP: Rutherford model predicts atoms are UNSTABLE (spiraling electron). This is why Bohr model was needed.**

3. Atomic Spectra

Hydrogen Spectrum — Balmer Formula (Empirical)

$$1/\lambda = R_H (1/n_1^2 - 1/n_2^2) \quad n_2 > n_1$$

where $R_H = 1.097 \times 10^7 \text{ m}^{-1}$ (Rydberg constant for hydrogen)

Spectral Series

Series	n_1	n_2	Region	First Line (approx.)
Lyman	1	2,3,4,...	UV	121.6 nm
Balmer	2	3,4,5,...	Visible/UV	656.3 nm (H-alpha)
Paschen	3	4,5,6,...	Near IR	1875 nm
Brackett	4	5,6,7,...	Mid IR	4051 nm
Pfund	5	6,7,8,...	Far IR	7460 nm

✓ **TRICK: Lyman=1, Balmer=2, Paschen=3, Brackett=4, Pfund=5. Memory: 'Lazy Bosses Pay Big Paychecks' — L,B,P,B,P.**

⚠ **TRAP: Lyman series is in UV, not visible. Only Balmer ($n_1=2$) series has lines in the visible region.**

4. Bohr Model

Bohr Postulates

- Electrons move in discrete circular orbits without radiating. (Stable stationary states.)
- Angular momentum of electron in n th orbit is quantized: $L = n\hbar = nh/2\pi$ ($n = 1,2,3,\dots$).
- Radiation is emitted or absorbed only when electron transitions between orbits: $h\nu = E_{\text{higher}} - E_{\text{lower}}$.

Derivation of Bohr Radius

Balance centripetal force with Coulomb attraction:

$$mv^2/r = ke^2/r^2 \quad \rightarrow \quad mvr = \sqrt{(ke^2mr)}$$

Apply angular momentum quantization: $mvr = n\hbar$

From above: $r = n^2\hbar^2/(mke^2)$

$$r_n = n^2 \times a_0 \quad \text{where } a_0 = \hbar^2/(mke^2) = 0.529 \text{ \AA} \text{ (Bohr radius)}$$

Derivation of Energy Levels

$$E_n = -13.6/n^2 \text{ eV} = -me^4/(8\epsilon_0^2 h^2 n^2)$$

Ground state ($n=1$): $E_1 = -13.6 \text{ eV}$. Ionization energy = 13.6 eV.

Bohr Velocity

$$v_n = ke^2/(n\hbar) = e^2/(4\pi\epsilon_0\hbar n) = v_1/n \quad \text{where } v_1 = 2.18 \times 10^6 \text{ m/s}$$

Bohr Frequency

$$v_n = v_n / (2\pi r_n) = mk^2 e^4 / (4\pi^2 \hbar^3) \times 1/n^3$$

Spectral Lines from Bohr

$$1/\lambda = R_H (1/n_1^2 - 1/n_2^2)$$

$$R_H = me^4 / (8\epsilon_0^2 h^3 c) = 1.097 \times 10^7 \text{ m}^{-1}$$

Key Values to Remember

n	r_n (Å)	E_n (eV)	v_n (m/s)
1	0.529	-13.6	2.18×10^6
2	2.116	-3.4	1.09×10^6
3	4.761	-1.51	0.727×10^6
∞	∞	0	0

✓ TRICK: $r_n = n^2 \times 0.529 \text{ Å}$. $E_n = -13.6/n^2 \text{ eV}$. These two formulas cover 80% of Bohr numericals.

⚠ TRAP: The energy E_n is NEGATIVE (bound state). Ionization energy = $|E_1| = 13.6 \text{ eV}$, NOT -13.6 eV .

5. Sommerfeld Model

Modification

Sommerfeld extended Bohr's model by allowing elliptical orbits in addition to circular ones. Two quantum numbers are introduced:

- Principal quantum number n : determines energy (same as Bohr).
- Azimuthal quantum number k ($= l+1$): determines shape of orbit ($1 \leq k \leq n$).

Key Result

$$E_n = -13.6/n^2 \text{ eV} \quad [\text{same as Bohr for H, no magnetic field}]$$

Sommerfeld model explains fine structure of spectral lines by incorporating relativistic corrections (different ellipses have different average radii and hence different relativistic energies).

Correspondence Principle

In the limit of large quantum numbers ($n \rightarrow \infty$), quantum mechanical results must agree with classical mechanics. Bohr used this to validate his model.

6. Reduced Mass Correction

Concept

In reality, both nucleus and electron revolve around their common center of mass. The reduced mass replaces electron mass:

$$\mu = m_e \times M / (m_e + M) \approx m_e (1 - m_e/M)$$

For hydrogen: $\mu_H = m_e \times m_p / (m_e + m_p) \approx 0.9995 m_e$

Effect on Rydberg Constant

$$R_X = R_\infty \times \mu/m_e \quad [\text{corrected Rydberg for element X}]$$

Explains why hydrogen (H) and deuterium (D) have very slightly different spectral lines.

7. LASER

Key Concepts

Stimulated Emission

When a photon of energy exactly equal to the energy difference between two levels ($h\nu = E_2 - E_1$) interacts with an atom in the excited state, it triggers the emission of a second photon. The emitted photon is identical to the incident photon (same frequency, phase, direction, polarization). This is the basis of laser action.

Spontaneous Emission

An atom in an excited state spontaneously falls to a lower state after a characteristic lifetime ($\sim 10^{-8}$ s) without any external trigger. Photon emitted in a random direction with random phase.

Population Inversion

Normally, higher energy levels have fewer atoms (Boltzmann distribution). For laser action, we need $N_2 > N_1$ (more atoms in upper level than lower) — this is population inversion. It is a non-equilibrium condition achieved by pumping.

Metastable State

An excited state with much longer lifetime ($\sim 10^{-3}$ s, versus $\sim 10^{-8}$ s for normal). Atoms accumulate in metastable states, enabling population inversion. Essential for laser operation.

Einstein A and B Coefficients

$$\text{Rate of stimulated emission} = B_{21} \times \rho(\nu) \times N_2$$

$$\text{Rate of spontaneous emission} = A_{21} \times N_2$$

$$\text{Rate of absorption} = B_{12} \times \rho(\nu) \times N_1$$

$$A_{21}/B_{21} = 8\pi h\nu^3/c^3 \quad [\text{Einstein relation}]$$

Laser Characteristics

Property	Description
Monochromaticity	Extremely narrow wavelength band
Coherence	High spatial and temporal coherence
Directionality	Very low divergence beam
High intensity	Enormous power density possible

Types of Laser

Type	Example	Medium	Wavelength
Solid-state	Ruby Laser	Cr^{3+} in Al_2O_3	694 nm (red)
Gas laser	He-Ne Laser	He + Ne gas	632.8 nm (red)

Type	Example	Medium	Wavelength
Semiconductor	Diode Laser	GaAs	800–900 nm (IR)

Ruby Laser (3-level system)

- Chromium ions pumped from ground E_1 to absorption band E_3 by green/blue light.
- Fast non-radiative decay from E_3 to metastable state E_2 .
- Population inversion achieved between E_2 and E_1 .
- Stimulated emission at 694 nm (red) when photon triggers $E_2 \rightarrow E_1$ transition.

He-Ne Laser (4-level system)

- He atoms excited to 20.61 eV by electric discharge.
- Resonant energy transfer from He to Ne atoms (excited to 20.66 eV level).
- Stimulated emission from Ne at 632.8 nm.
- Ne ground-state atoms pumped away by collisions with tube walls.

Laser Applications

- Medicine: eye surgery (LASIK), cancer treatment, cutting.
- Communications: fiber optic networks.
- Industry: cutting, welding, drilling.
- Defense: range finders, guidance systems.
- Research: spectroscopy, holography.

Unit II — Exam Question Bank

2-Mark Questions

- What is population inversion? Why is it needed for laser action?
- Distinguish spontaneous and stimulated emission.
- What is a metastable state?
- State Bohr's second postulate.
- What is the ionization energy of hydrogen?

5-Mark Questions

- State the failures of Rutherford's model. How did Bohr overcome them?
- Derive Bohr's formula for energy levels.
- Explain population inversion and metastable states. Why can't a 2-level laser work?
- List and explain the spectral series of hydrogen.

10-Mark Questions

- Describe Rutherford's alpha scattering experiment. State observations, conclusions, and failures.
- Derive Bohr radius, velocity, and energy levels from first principles. Hence derive the Rydberg formula.
- Explain the construction and working of a ruby laser (or He-Ne laser). Draw energy level diagrams.

Common Mistakes

- ⚠ **TRAP:** Forgetting that in Rutherford's formula, the scattering depends on $\sin^4(\theta/2)$, NOT $\sin(\theta)$.
- ⚠ **TRAP:** The Rydberg constant is $R = 1.097 \times 10^7 \text{ m}^{-1}$ — notice it's per METER.
- ⚠ **TRAP:** In 2-level systems, you cannot achieve population inversion because stimulated emission and absorption have equal probability; 3 or 4 levels are required.

UNIT III — QUANTUM MECHANICS

1. Wave Function (ψ)

Definition

The wave function $\psi(x,t)$ is a mathematical function that completely describes the quantum state of a particle. It contains all possible information about the system.

Physical Interpretation (Born's Rule)

$$|\psi(x,t)|^2 dx = \text{Probability of finding particle between } x \text{ and } x+dx$$

$|\psi|^2 = \psi^*\psi$ is the probability density. This is the 'probabilistic interpretation' by Max Born.

Normalization Condition

$$\int_{-\infty}^{+\infty} |\psi(x,t)|^2 dx = 1$$

The total probability of finding the particle somewhere in space is 1 (certainty).

Well-Behaved Wave Function (Conditions)

21. ψ must be single-valued (one probability at each point).
22. ψ must be finite everywhere (probability must be bounded).
23. ψ must be continuous (no jumps).
24. $d\psi/dx$ must be continuous (except where potential is infinite).
25. $\psi \rightarrow 0$ as $x \rightarrow \pm\infty$ (particle must be localized).

⚠ TRAP: A wave function that doesn't satisfy these conditions is physically meaningless. Exam often asks you to test a given ψ .

2. Expectation Values

Position Expectation Value

$$\langle x \rangle = \int_{-\infty}^{+\infty} \psi^* x \psi dx$$

Momentum Expectation Value

$$\langle p \rangle = \int_{-\infty}^{+\infty} \psi^* (-i\hbar \partial/\partial x) \psi dx$$

Energy Expectation Value

$$\langle E \rangle = \int_{-\infty}^{+\infty} \psi^* \hat{H} \psi dx$$

General Expectation Value

$$\langle \hat{Q} \rangle = \int \psi^* \hat{Q} \psi dx$$

The operator acts on ψ (to the right), then multiply by ψ^* , then integrate. Order matters!

3. Schrödinger Equation

Time-Dependent Schrödinger Equation (TDSE)

$$i\hbar \partial\psi/\partial t = \hat{H}\psi = \left[-\hbar^2/2m \partial^2/\partial x^2 + V(x, t)\right] \psi$$

Derivation Outline

26. Start with de Broglie: $\lambda = h/p$, so $k = p/\hbar$.
27. For free particle: $\psi = A e^{i(kx - \omega t)} = A e^{i(px - Et)/\hbar}$.
28. Note: $\partial\psi/\partial t = (-iE/\hbar)\psi \rightarrow i\hbar\partial\psi/\partial t = E\psi$.
29. Note: $\partial^2\psi/\partial x^2 = (-p^2/\hbar^2)\psi \rightarrow -\hbar^2/2m \partial^2\psi/\partial x^2 = p^2/2m \psi = KE \cdot \psi$.
30. Since $E = KE + V$: $i\hbar\partial\psi/\partial t = (-\hbar^2/2m \partial^2/\partial x^2 + V)\psi$.

Time-Independent Schrödinger Equation (TISE)

For stationary states, $\psi(x, t) = \phi(x)e^{-iEt/\hbar}$. Substituting into TDSE:

$$\hat{H}\phi = E\phi \rightarrow \left[-\hbar^2/2m d^2/dx^2 + V(x)\right] \phi(x) = E \phi(x)$$

$$d^2\phi/dx^2 + (2m/\hbar^2)(E - V)\phi = 0$$

Operators

Observable	Classical	Quantum Operator
Position	x	$\hat{x} = x$ (multiply by x)
Momentum	p	$\hat{p} = -i\hbar \partial/\partial x$
Kinetic Energy	$p^2/2m$	$\hat{K} = -\hbar^2/2m \partial^2/\partial x^2$
Potential Energy	$V(x)$	$\hat{V} = V(x)$
Total Energy (Hamiltonian)	$E = KE + PE$	$\hat{H} = -\hbar^2/2m \partial^2/\partial x^2 + V$

Eigenfunctions and Eigenvalues

An eigenfunction ϕ of operator $\hat{Q}$ satisfies:

$$\hat{Q} \phi = q \phi \quad [q \text{ is the eigenvalue (measured value)}]$$

Example: $\hat{p} e^{ikx} = -i\hbar(ik)e^{ikx} = \hbar k e^{ikx} \rightarrow \text{eigenvalue} = \hbar k = p$

Superposition Principle

If ϕ_1 and ϕ_2 are solutions of Schrödinger equation, then $a\phi_1 + b\phi_2$ is also a solution (Linearity). Any quantum state can be written as a superposition of eigenstates.

Unit III — Exam Question Bank

2-Mark Questions

- What is the physical interpretation of $|\psi|^2$?
- State the normalization condition.
- What is an eigenfunction? Give an example.
- Write the time-independent Schrödinger equation.

- What are the conditions for a well-behaved wave function?

5-Mark Questions

- Derive the time-dependent Schrödinger equation.
- Define expectation values. Write formulas for $\langle x \rangle$ and $\langle p \rangle$.
- Explain the physical meaning of the wave function. State Born's rule.

10-Mark Questions

- Derive the time-dependent Schrödinger equation. Hence obtain the time-independent form for stationary states.
- Define wave function, normalization, and expectation values. Explain what physical information ψ contains.

Common Mistakes

- ⚠ **TRAP:** The operator acts **BEFORE** multiplying by ψ^* . Writing $\int \psi^* (\hat{p}\psi) dx \neq \int (\hat{p}\psi^*) \psi dx$.
- ⚠ **TRAP:** The TISE is for stationary states **ONLY**. For time-varying potentials, use the full TDSE.

UNIT IV — APPLICATIONS OF SCHRÖDINGER EQUATION

1. Free Particle

Setup

$V(x) = 0$ everywhere. The TISE becomes:

$$d^2\psi/dx^2 + k^2\psi = 0 \quad \text{where } k^2 = 2mE/\hbar^2 \rightarrow k = \sqrt{(2mE)}/\hbar$$

Solution

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \quad [\text{traveling waves}]$$

Energy eigenvalues: $E = \hbar^2 k^2 / 2m$ (continuous spectrum — any positive E is allowed).

Free particle has a continuous energy spectrum (no quantization) because it is not confined. Confinement $\rightarrow$ quantization.

2. Particle in an Infinite Potential Well (Particle in a Box)

Setup

$V(x) = 0$ for $0 < x < L$, $V(x) = \infty$ for $x \leq 0$ and $x \geq L$.

Particle is strictly confined to $[0, L]$. This is the most important problem in quantum mechanics.

Boundary Conditions

$$\psi(0) = 0 \quad \text{and} \quad \psi(L) = 0 \quad [\text{wave function must vanish at infinite walls}]$$

Derivation — Full

31. TISE inside well: $d^2\psi/dx^2 + k^2\psi = 0$, $k = \sqrt{(2mE)}/\hbar$

32. General solution: $\psi(x) = A \sin(kx) + B \cos(kx)$

33. Apply $\psi(0) = 0$: $B = 0 \rightarrow \psi(x) = A \sin(kx)$

34. Apply $\psi(L) = 0$: $A \sin(kL) = 0 \rightarrow kL = n\pi \rightarrow k_n = n\pi/L$

35. Energy eigenvalues: $E_n = \hbar^2 k_n^2 / 2m = n^2 \pi^2 \hbar^2 / (2mL^2) = n^2 h^2 / (8mL^2)$

$$E_n = n^2 E_1 \quad \text{where } E_1 = \pi^2 \hbar^2 / (2mL^2) = h^2 / (8mL^2) \quad [\text{Ground state energy}]$$

$$E_n = n^2 \times (h^2 / 8mL^2) \quad n = 1, 2, 3, \dots$$

36. Normalize: $\int_0^L A^2 \sin^2(n\pi x/L) dx = 1 \rightarrow A = \sqrt{(2/L)}$

$$\psi_n(x) = \sqrt{(2/L)} \sin(n\pi x/L) \quad \text{for } 0 < x < L, = 0 \text{ otherwise}$$

Key Results

n	$\psi_n(x)$	E_n	Nodes (inside well)
1 (ground)	$\sqrt{(2/L)} \sin(\pi x/L)$	E_1	0
2 (1st excited)	$\sqrt{(2/L)} \sin(2\pi x/L)$	$4E_1$	1 (at $L/2$)
3 (2nd)	$\sqrt{(2/L)} \sin(3\pi x/L)$	$9E_1$	2 (at $L/3, 2L/3$)

n	$\psi_n(x)$	E_n	Nodes (inside well)
excited)			
n	$\sqrt{2/L} \sin(n\pi x/L)$	$n^2 E_1$	$n-1$

Expectation Values in a Box

$\langle x \rangle = L/2$ [particle equally likely left or right – symmetric]

$$\langle x^2 \rangle = L^2/3 - L^2/(2n^2\pi^2)$$

$\langle p \rangle = 0$ [equal probability of +p and -p]

$$\langle p^2 \rangle = (n\pi\hbar/L)^2 = 2mE_n$$

$$\langle E \rangle = E_n = n^2\hbar^2/(8mL^2)$$

✓ TRICK: Ground state energy $E_1 = \hbar^2/(8mL^2)$. For electron in 1 Å box: $E_1 \approx 37.6$ eV. For proton in 1 fm box: $E_1 \approx 20$ MeV.

⚠ TRAP: $n=0$ is NOT allowed — would give $\psi=0$ everywhere (no particle!). Minimum is $n=1$ (zero-point energy).

3. Finite Potential Well and Tunneling

Finite Well

$V(x) = 0$ for $-L/2 < x < L/2$, $V(x) = V_0$ for $|x| > L/2$.

Unlike infinite well: wave function penetrates into classically forbidden regions ($|x| > L/2$), decaying exponentially.

$$\psi \sim e^{\pm\kappa x} \text{ outside, where } \kappa = \sqrt{2m(V_0 - E)}/\hbar$$

Tunnel Effect (Quantum Tunneling)

A quantum particle can penetrate through a potential barrier even if its energy E is less than the barrier height V_0 . This is classically impossible but quantum-mechanically allowed because the wave function has a non-zero (though small) amplitude inside and beyond the barrier.

Transmission Probability (WKB Approximation)

$$T \approx e^{-2\kappa a} \quad \text{where } \kappa = \sqrt{2m(V_0 - E)}/\hbar, \quad a = \text{barrier width}$$

$$\text{More precisely: } T = e^{-2\int \kappa(x) dx} \quad [\text{for slowly varying barrier}]$$

Applications of Tunneling

- Alpha decay: alpha particle tunnels through nuclear Coulomb barrier.
- Scanning Tunneling Microscope (STM): tunneling current images atoms.
- Tunnel diode, Josephson junction.

Scanning Tunneling Microscope (STM)

A sharp metallic tip is brought within 1 nm of a conducting surface. Quantum tunneling of electrons between tip and surface produces a current:

$$I \propto e^{-2\kappa d} \quad [\text{exponentially sensitive to tip-surface distance } d]$$

A change of just 1 Å in d changes the current by about an order of magnitude! This extreme sensitivity allows atomic-resolution imaging.

4. Quantum Harmonic Oscillator

Potential

$$V(x) = \frac{1}{2} m \omega^2 x^2 \quad [\text{parabolic potential}]$$

TISE for Harmonic Oscillator

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2} m \omega^2 x^2 \psi = E \psi$$

Energy Eigenvalues

$$E_n = (n + \frac{1}{2}) \hbar \omega \quad n = 0, 1, 2, 3, \dots$$

Zero-Point Energy

$$E_0 = \frac{1}{2} \hbar \omega \quad [n=0, \text{ground state} - \text{NOT zero!}]$$

The zero-point energy $\frac{1}{2} \hbar \omega$ exists because of the Heisenberg uncertainty principle: the particle cannot be at rest at $x=0$ (that would require $\Delta x=0$ and $\Delta p=0$ simultaneously).

Comparison with PIB

Property	Particle in Box	Harmonic Oscillator
Potential	$V=0$ inside, ∞ outside	$V = \frac{1}{2} m \omega^2 x^2$
Energy spacing	Increases with n ($\propto n^2$)	Constant ($\hbar \omega$)
Ground state E	$h^2/8mL^2$	$\frac{1}{2} \hbar \omega$
Wave function	sin functions (confined)	Gaussian $\times$ Hermite poly.
Zero-point energy	Yes ($n=0$ not allowed)	$\frac{1}{2} \hbar \omega$

Unit IV — Exam Question Bank

2-Mark Questions

- What is zero-point energy? Why does it exist?
- State the boundary conditions for particle in a box.
- What is quantum tunneling?
- Write the energy eigenvalues for harmonic oscillator.

5-Mark Questions

- Derive energy eigenvalues for particle in infinite potential well.
- Explain tunneling effect and its applications (alpha decay, STM).
- Why can't the ground state energy of a particle in a box be zero?

10-Mark Questions

- Completely solve the particle in an infinite potential well: find eigenfunctions, eigenvalues, and show normalization.

- Derive the energy levels of quantum harmonic oscillator. Explain zero-point energy using Heisenberg uncertainty principle.

Common Mistakes

- ⚠ **TRAP:** Students write $E = n^2 h^2 / 8mL^2$ then use $\hbar$ (h-bar) instead of h — use h (not $\hbar$) in this formula!
- ⚠ **TRAP:** The tunneling probability $T \approx e^{-2\kappa a}$ is an APPROXIMATION for a square barrier. Always state the approximation.
- ⚠ **TRAP:** In harmonic oscillator, n starts from 0 (not 1). Ground state is $n=0$.

UNIT V — QUANTUM COMPUTING & QUANTUM CORRELATIONS

1. Classical Bit vs Qubit

Property	Classical Bit	Qubit
States	0 or 1 only	$ 0\rangle, 1\rangle$, or superposition $\alpha 0\rangle + \beta 1\rangle$
Measurement	0 or 1 (definite)	0 with prob $ \alpha ^2$, 1 with prob $ \beta ^2$
Copying	Freely copyable	No-cloning theorem: cannot copy
Parallelism	One value at a time	Superposition of many values
Physical impl.	Transistor, capacitor	Spin, photon polarization, etc.

2. Qubit and Dirac (Bra-Ket) Notation

Ket and Bra

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad [\text{ket} = \text{state vector}]$$
$$\langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1| \quad [\text{bra} = \text{Hermitian conjugate}]$$
$$|\alpha|^2 + |\beta|^2 = 1 \quad [\text{normalization}]$$

Measurement Probabilities

$$P(0) = |\alpha|^2 = |\langle 0|\psi\rangle|^2$$
$$P(1) = |\beta|^2 = |\langle 1|\psi\rangle|^2$$

Standard Basis States

$$|0\rangle = [1, 0]^T \quad |1\rangle = [0, 1]^T$$

Superposition States

$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2} \quad [\text{equal superposition, prob 50:50}]$$
$$|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

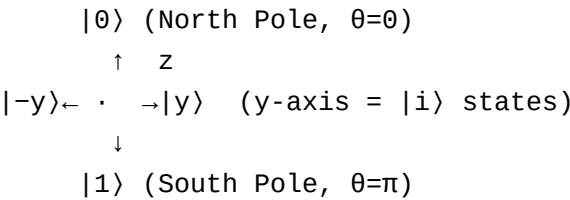
3. Bloch Sphere

Any single qubit state can be written as:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle$$

where $\theta \in [0,\pi]$ is the polar angle and $\varphi \in [0,2\pi]$ is the azimuthal angle. This maps any qubit to a point on the surface of a unit sphere (the Bloch sphere).

Bloch Sphere (ASCII Diagram):



x-axis: $|+\rangle$ ($\varphi=0$) and $|-\rangle$ ($\varphi=\pi$)

State	Bloch Sphere Position	θ	φ
$ 0\rangle$	North Pole	0	–
$ 1\rangle$	South Pole	π	–
$ +\rangle=(0\rangle+ 1\rangle)/\sqrt{2}$	+x axis	$\pi/2$	0
$ -\rangle=(0\rangle- 1\rangle)/\sqrt{2}$	-x axis	$\pi/2$	π
$ i\rangle=(0\rangle+i 1\rangle)/\sqrt{2}$	+y axis	$\pi/2$	$\pi/2$

4. Bell States (Maximally Entangled Two-Qubit States)

The Four Bell States

$$\begin{aligned} |\Phi^+\rangle &= (|00\rangle + |11\rangle)/\sqrt{2} \\ |\Phi^-\rangle &= (|00\rangle - |11\rangle)/\sqrt{2} \\ |\Psi^+\rangle &= (|01\rangle + |10\rangle)/\sqrt{2} \\ |\Psi^-\rangle &= (|01\rangle - |10\rangle)/\sqrt{2} \end{aligned}$$

Bell states are the maximally entangled two-qubit states. They form an orthonormal basis for the 4-dimensional two-qubit Hilbert space.

Properties of Bell States

- Measurement of one qubit instantly determines the state of the other, regardless of distance.
- Cannot be written as product states $A\otimes B$ (non-separable = entangled).
- Violate Bell inequalities ($S_{\text{max}} = 2\sqrt{2} > 2$, quantum limit > classical limit).

5. Quantum Entanglement and Bell Inequalities

Entanglement

Two qubits are entangled if their joint quantum state CANNOT be written as a product of individual qubit states:

$$|\psi\rangle_{AB} \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

Measuring one qubit affects the other instantaneously, regardless of separation (Einstein called this 'spooky action at a distance').

Bell Inequalities (CHSH Inequality)

$$\begin{aligned} S &= |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \\ \text{Classical (local hidden variable): } S &\leq 2 \\ \text{Quantum mechanics: } S &\leq 2\sqrt{2} \approx 2.828 \text{ (Tsirelson bound)} \end{aligned}$$

Experiments (Aspect, 1982; Clauser) show $S > 2$, confirming quantum mechanics violates Bell inequalities — no local hidden variable theory can explain quantum entanglement.

Schmidt Decomposition

Any bipartite pure state $|\psi\rangle_{AB}$ can be written as:

$$|\psi\rangle_{AB} = \sum_i \lambda_i |a_i\rangle |b_i\rangle \quad [\text{Schmidt decomposition}]$$

where $\lambda_i \geq 0$ are Schmidt coefficients with $\sum \lambda_i^2 = 1$, and $\{|a_i\rangle\}$, $\{|b_i\rangle\}$ are orthonormal bases.

Schmidt rank = number of non-zero λ_i values

- Schmidt rank = 1: state is SEPARABLE (not entangled).
- Schmidt rank > 1: state is ENTANGLED.

6. Quantum Gates and Quantum Correlations

Key Single-Qubit Gates

Gate	Symbol	Matrix	Action
Pauli X (NOT)	X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$ 0\rangle \leftrightarrow 1\rangle$ (bit flip)
Pauli Y	Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	Phase + bit flip
Pauli Z	Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$ 1\rangle \rightarrow - 1\rangle$ (phase flip)
Hadamard	H	$1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$ 0\rangle \rightarrow +\rangle$, $ 1\rangle \rightarrow -\rangle$
Phase (S)	S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	Phase $\pi/2$

7. Superdense Coding

Protocol

Superdense coding allows Alice to send 2 classical bits to Bob using only 1 qubit (plus pre-shared entanglement). This is a 2:1 quantum advantage.

Steps

37. Alice and Bob share a Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ (pre-shared).

38. Alice wants to send 2 classical bits (00, 01, 10, or 11). She applies a gate to HER qubit only:

Alice sends	Gate applied	Resulting Bell state
00	I (do nothing)	$ \Phi^+\rangle = (00\rangle + 11\rangle)/\sqrt{2}$
01	X (bit flip)	$ \Psi^+\rangle = (01\rangle + 10\rangle)/\sqrt{2}$
10	Z (phase flip)	$ \Phi^-\rangle = (00\rangle - 11\rangle)/\sqrt{2}$
11	iY = ZX	$ \Psi^-\rangle = (01\rangle - 10\rangle)/\sqrt{2}$

39. 3. Alice sends her qubit to Bob.

40. 4. Bob performs a Bell measurement (CNOT + Hadamard) to determine which of the 4 Bell states he has, recovering both bits.

Superdense coding: 1 qubit + 1 ebit (entanglement) = 2 classical bits of information.

8. Quantum Teleportation

Purpose

Teleportation transfers an unknown quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from Alice to Bob using a pre-shared Bell pair and 2 classical bits. The original state at Alice's location is destroyed (no-cloning theorem is not violated).

Protocol Steps

- 41. Alice has: $|\psi\rangle_1 = \alpha|0\rangle + \beta|1\rangle$ (unknown state to teleport).
- 42. Alice and Bob share Bell pair $|\Phi^+\rangle_{23} = (|00\rangle + |11\rangle)/\sqrt{2}$.
- 43. Total system: $|\psi\rangle_1 \otimes |\Phi^+\rangle_{23}$.
- 44. Alice applies CNOT (control=1, target=2), then Hadamard on qubit 1.
- 45. Alice measures qubits 1 and 2 → gets classical bits (00, 01, 10, or 11).
- 46. Alice sends 2 classical bits to Bob.
- 47. Bob applies correction gate based on Alice's bits: 00 → I, 01 → X, 10 → Z, 11 → ZX.
- 48. Bob's qubit 3 is now $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Teleportation complete!

Alice's measurement	Bob's correction gate	Meaning
00	I (nothing)	State already correct
01	X (bit flip)	Correct bit flip
10	Z (phase flip)	Correct phase flip
11	ZX (both)	Correct both

Key constraint: 2 classical bits MUST be sent from Alice to Bob. Therefore teleportation CANNOT transmit information faster than light (no FTL signaling).

⚠ TRAP: Teleportation does NOT copy the state — Alice's original qubit is destroyed upon measurement. No-cloning is safe.

Comparison: Superdense Coding vs Teleportation

Feature	Superdense Coding	Quantum Teleportation
Direction	2 cbits → 1 qubit	1 qubit → 2 cbits
Input	2 classical bits	1 unknown qubit state
Output	Bob learns 2 cbits	Bob gets the qubit state
Resource	1 qubit sent + 1 ebit	2 cbits sent + 1 ebit
Original preserved?	N/A	No (destroyed by measurement)

Unit V — Exam Question Bank

2-Mark Questions

- What is a qubit? How does it differ from a classical bit?
- What is quantum entanglement?
- Write the four Bell states.
- What is the Bloch sphere representation?

- State Bell's inequality. What is its significance?

5-Mark Questions

- Explain Dirac notation. Define bra, ket, inner product, and normalization.
- Explain quantum entanglement and Bell states. Why do Bell states violate Bell inequalities?
- Explain Schmidt decomposition and Schmidt rank.

10-Mark Questions

- Explain the protocol of quantum teleportation step-by-step. Draw the circuit diagram. What is the role of classical communication?
- Explain superdense coding. Show how 2 classical bits are encoded and decoded using 1 qubit and a Bell pair.
- Compare quantum computing with classical computing. Explain qubits, superposition, entanglement, and Bloch sphere.

Common Mistakes

- ⚠ **TRAP: Teleportation does NOT violate no-cloning—Alice's state is DESTROYED on measurement.**
- ⚠ **TRAP: Teleportation is NOT faster-than-light: 2 classical bits must still be sent via normal channel.**
- ⚠ **TRAP: Superdense coding sends 2 classical bits using 1 qubit, NOT 2 qubits.**

SECTION A — MASTER FORMULA SHEET

Unit I Formulas

Formula	Equation	Notes
Planck energy quantum	$E = h\nu = hc/\lambda$	$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
Stefan-Boltzmann	$E = \sigma T^4$	$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Wien displacement	$\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m}\cdot\text{K}$	—
Planck radiation law	$u(\nu) = (8\pi h \nu^3 / c^3) / (e^{\{h\nu/kT\}} - 1)$	—
Einstein photoelectric	$KE_{\text{max}} = h\nu - \phi$	—
Stopping potential	$eV_s = h\nu - \phi$	—
Threshold wavelength	$\lambda_0 = hc/\phi$	—
Quick $h\nu$ in eV	$h\nu(\text{eV}) = 12400/\lambda(\text{\AA})$	Shortcut
Compton shift	$\Delta\lambda = (h/m_e c)(1 - \cos\theta)$	—
Compton wavelength	$\lambda_C = h/m_e c = 0.02426 \text{ \AA}$	—
Pair production min E	$h\nu_{\text{min}} = 2m_e c^2 = 1.022 \text{ MeV}$	—
De Broglie wavelength	$\lambda = h/p = h/mv$	—
λ for electrons (quick)	$\lambda(\text{\AA}) = 12.26/\sqrt{V}$	V in volts
λ from KE	$\lambda = h/\sqrt{2mE_K}$	—
Phase velocity	$v_p = \omega/k = E/p$	—
Group velocity	$v_g = d\omega/dk = v_{\text{particle}}$	—
Uncertainty (position-momentum)	$\Delta x \cdot \Delta p \geq \hbar$	—
Uncertainty (energy-time)	$\Delta E \cdot \Delta t \geq \hbar$	—

Unit II Formulas

Formula	Equation	Notes
Rydberg formula	$1/\lambda = R(1/n_1^2 - 1/n_2^2)$	$R = 1.097 \times 10^7 \text{ m}^{-1}$
Bohr radius	$r_n = n^2 a_0$	$a_0 = 0.529 \text{ \AA}$
Bohr energy	$E_n = -13.6/n^2 \text{ eV}$	—
Bohr velocity	$v_n = v_1/n$	$v_1 = 2.18 \times 10^6 \text{ m/s}$
Distance of closest approach	$r_{\text{min}} = Z_1 Z_2 k e^2 / E_K$	—
Einstein A/B relation	$A_{21}/B_{21} = 8\pi h \nu^3 / c^3$	—
Reduced mass	$\mu = m_e M / (m_e + M)$	—

Unit III Formulas

Formula	Equation	Notes
Normalization	$\int \psi ^2 dx = 1$	—
Probability density	$P(x) = \psi(x) ^2$	—
Position expectation	$\langle x \rangle = \int \psi^* x \psi dx$	—
Momentum expectation	$\langle p \rangle = \int \psi^* (-i\hbar \partial/\partial x) \psi dx$	—
TDSE	$i\hbar \partial \psi / \partial t = \hat{H} \psi$	—
TISE	$\hat{H} \psi = E \psi$	Stationary states
Momentum operator	$\hat{p} = -i\hbar \partial/\partial x$	—
Hamiltonian	$\hat{H} = -\hbar^2/2m \nabla^2 + V$	—
Eigenvalue equation	$\hat{Q} \psi = q \psi$	q = measured value

Unit IV Formulas

Formula	Equation	Notes
PIB eigenfunctions	$\psi_n = \sqrt{2/L} \sin(n\pi x/L)$	—
PIB energy levels	$E_n = n^2 \hbar^2 / (8mL^2)$	Use h , not $\hbar$
PIB ground state	$E_1 = \hbar^2 / 8mL^2$	—
PIB $\langle x \rangle$	$\langle x \rangle = L/2$	—
PIB $\langle p \rangle$	$\langle p \rangle = 0$	—
Tunneling decay	$\kappa = \sqrt{2m(V_0 - E)} / \hbar$	—
Tunneling probability	$T \approx e^{-2\kappa a}$	Square barrier WKB
HO energy levels	$E_n = (n + 1/2) \hbar \omega$	$n = 0, 1, 2, \dots$
HO zero-point energy	$E_0 = 1/2 \hbar \omega$	—

Unit V Formulas

Formula	Equation	Notes
Qubit state	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$	$ \alpha ^2 + \beta ^2 = 1$
Bloch sphere	$ \psi\rangle = \cos(\theta/2) 0\rangle + e^{i\phi} \sin(\theta/2) 1\rangle$	$\theta \in [0, \pi], \phi \in [0, 2\pi)$
Bell state Φ^+	$(00\rangle + 11\rangle) / \sqrt{2}$	—
Bell state Φ^-	$(00\rangle - 11\rangle) / \sqrt{2}$	—
Bell state Ψ^+	$(01\rangle + 10\rangle) / \sqrt{2}$	—
Bell state Ψ^-	$(01\rangle - 10\rangle) / \sqrt{2}$	—
CHSH classical bound	$S \leq 2$	Local hidden variable
CHSH quantum bound	$S \leq 2\sqrt{2} \approx 2.828$	Tsirelson bound

SECTION B — DERIVATIONS TO MEMORIZE

Unit I Derivations

49. Compton shift formula $\Delta\lambda = (h/m_e c)(1 - \cos\theta)$: Use energy + x-momentum + y-momentum conservation, eliminate electron angle ϕ .
50. Group velocity equals particle velocity: $v_g = d\omega/dk = dE/dp = p/m = v$.
51. Heisenberg uncertainty from wave packet: $\Delta k \cdot \Delta x \geq 1$ (Fourier), then $\Delta p = \hbar \Delta k$.

Unit II Derivations

52. Bohr radius: Force balance ($mv^2/r = ke^2/r^2$) + angular momentum quantization ($mvr = n\hbar$).
53. Bohr energy levels: $E_n = KE + PE = \frac{1}{2}mv^2 - ke^2/r = -ke^2/2r = -13.6/n^2 \text{ eV}$.
54. Rydberg formula from Bohr: $h\nu = E_{n2} - E_{n1}$, then $\nu = c/\lambda$.

Unit III Derivations

55. Time-dependent Schrödinger equation: Start with $\psi = e^{i(kx - \omega t)}$, apply energy/momentum operators.
56. Time-independent Schrödinger equation: Separate $\psi(x, t) = \phi(x)T(t)$, use TDSE.

Unit IV Derivations

57. Particle in a box eigenfunctions: Apply boundary conditions to general solution $A \sin(kx) + B \cos(kx)$.
58. Particle in a box energy: $kL = n\pi \rightarrow k_n = n\pi/L \rightarrow E_n = \hbar^2 k^2/2m = n^2 \hbar^2/8mL^2$.
59. Normalization of PIB: $\int_0^L A^2 \sin^2(n\pi x/L) dx = 1 \rightarrow A = \sqrt{2/L}$.
60. Zero-point energy using HUP: $\Delta x \Delta p \geq \hbar$, $p \geq \hbar/L$, $E \geq \hbar^2/2mL^2 = h^2/8mL^2$.

Unit V Derivations

61. Show Bell state is entangled: attempt to write $(|00\rangle + |11\rangle)/\sqrt{2} = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$, show contradiction.
62. Teleportation circuit: Start from $|\psi\rangle_1 |\Phi^+\rangle_{23}$, apply CNOT, H, measure, show Bob's state.

SECTION C — TOP 50 MOST IMPORTANT EXAM QUESTIONS

#	Question	Unit	Type
1	Derive Planck's radiation law and show the Wien and Rayleigh-Jeans limits.	I	10M
2	Explain the photoelectric effect experiments. State Einstein's equation and derive stopping potential.	I	10M
3	Derive the Compton shift formula using conservation of energy and momentum.	I	10M
4	Explain pair production and pair annihilation with energy conditions.	I	5M
5	State de Broglie hypothesis. Derive de Broglie wavelength for electron accelerated through V volts.	I	5M
6	Distinguish phase velocity and group velocity. Prove group velocity equals particle velocity.	I	5M
7	Describe the Davisson-Germer experiment and show how it confirms de Broglie hypothesis.	I	10M
8	State and explain Heisenberg uncertainty principle. Give two applications.	I	10M
9	What is the ultraviolet catastrophe? How did Planck resolve it?	I	5M
10	Calculate de Broglie wavelength of electron with given energy/voltage.	I	Num
11	Describe Rutherford alpha scattering experiment. State results and limitations.	II	10M
12	State Bohr postulates. Derive Bohr radius and energy levels.	II	10M
13	From Bohr model, derive the Rydberg formula for hydrogen spectral lines.	II	10M
14	List all hydrogen spectral series. Give n values, spectral region.	II	5M
15	Find wavelength of first/last line of given series.	II	Num
16	What is reduced mass? How does it correct Rydberg constant?	II	5M
17	Explain Sommerfeld model. What improvement over Bohr?	II	5M
18	Distinguish spontaneous and stimulated emission. Define Einstein A, B coefficients.	II	5M
19	What is population inversion? Why can't a two-level system produce laser action?	II	5M
20	Explain construction and working of ruby laser (or He-Ne laser) with energy diagram.	II	10M

#	Question	Unit	Type
2 1	State physical significance of wave function. State Born's interpretation.	III	5M
2 2	What conditions must a wave function satisfy? (Well-behaved wave function)	III	5M
2 3	Write formulas for $\langle x \rangle$, $\langle p \rangle$, $\langle E \rangle$. Explain expectation values.	III	5M
2 4	Derive the time-dependent Schrödinger equation.	III	10M
2 5	Obtain time-independent Schrödinger equation from TDSE.	III	5M
2 6	What are eigenvalues and eigenfunctions? Give an example.	III	5M
2 7	State linearity and superposition principle in quantum mechanics.	III	2M
2 8	Write momentum and Hamiltonian operators explicitly.	III	2M
2 9	Solve the particle in an infinite potential well completely.	IV	10M
3 0	Show that ground state energy cannot be zero (use HUP argument).	IV	5M
3 1	Find the expectation values of x , p , and E for particle in a box.	IV	5M
3 2	What is quantum tunneling? Derive tunneling probability.	IV	5M
3 3	Explain alpha decay in terms of tunneling effect.	IV	5M
3 4	Explain how STM works using quantum tunneling.	IV	5M
3 5	What are the energy eigenvalues of harmonic oscillator? Explain zero-point energy.	IV	10M
3 6	Compare particle in a box with harmonic oscillator.	IV	5M
3 7	What is a qubit? Write qubit state in general form.	V	2M
3 8	Explain Bloch sphere representation. Draw and label.	V	5M
3 9	Write and explain the four Bell states. Prove one is entangled.	V	10M
4 0	State and explain Bell inequalities and CHSH inequality.	V	5M
4 1	What is the Schmidt decomposition? Define Schmidt rank.	V	5M
4 2	Explain quantum entanglement. What is 'spooky action at a distance'?	V	5M

#	Question	Unit	Type
4 3	Compare superdense coding and quantum teleportation.	V	5M
4 4	Explain superdense coding protocol step by step.	V	10M
4 5	Explain quantum teleportation protocol step by step. Why are classical bits needed?	V	10M
4 6	What are properties of quantum computing over classical computing?	V	5M
4 7	Explain Dirac (bra-ket) notation. Define inner product, normalization.	V	5M
4 8	What is the no-cloning theorem? Why does teleportation not violate it?	V	5M
4 9	Find stopping potential given frequency and work function.	I	Num
5 0	Find Compton shift at $\theta = 90^\circ$ and $\theta = 180^\circ$.	I	Num

SECTION D — COMMON NUMERICAL PATTERNS

Pattern 1: Blackbody / Planck

Find λ_{max} given T

Formula: $\lambda_{\text{max}} = b/T = 2.898 \times 10^{-3}/T$ (m)

Example: $T = 5800 \text{ K} \rightarrow \lambda_{\text{max}} = 2.898 \times 10^{-3}/5800 = 499.6 \text{ nm}$ (visible — yellow-green)

Pattern 2: Photoelectric Effect

Find stopping potential V_s

1. Find $h\nu$: use $h\nu(\text{eV}) = 12400/\lambda(\text{\AA})$ or $1240/\lambda(\text{nm})$

2. $KE_{\text{max}} = h\nu - \phi$ (both in eV)

3. $V_s = KE_{\text{max}}$ (in volts)

Pattern 3: Compton Effect

Find $\Delta\lambda$, λ' , and recoil energy of electron

1. $\Delta\lambda = 0.02426(1 - \cos\theta) \text{ \AA}$

2. $\lambda' = \lambda + \Delta\lambda$

3. $h\nu = hc/\lambda$, $h\nu' = hc/\lambda'$, $\Delta E_{\text{photon}} = h\nu - h\nu'$

4. Recoil KE of electron = $h\nu - h\nu' = \Delta E_{\text{photon}}$

Pattern 4: De Broglie Wavelength

Electron accelerated through V volts

Quick formula: $\lambda(\text{\AA}) = 12.26/\sqrt{V}$ (V in volts)

From first principles: $\lambda = h/\sqrt{(2meV)} = h/\sqrt{(2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19} \times V)}$

Pattern 5: Heisenberg Uncertainty

Find minimum KE given confinement size Δx

1. $\Delta p = \hbar/\Delta x$

2. $KE_{\text{min}} = (\Delta p)^2/2m = \hbar^2/(2m\Delta x^2)$

Pattern 6: Bohr Model

Find wavelength of transition $n_2 \rightarrow n_1$

$1/\lambda = R(1/n_1^2 - 1/n_2^2)$

For first line of Lyman: $n_1=1$, $n_2=2 \rightarrow 1/\lambda = R(1 - 1/4) = 3R/4 \rightarrow \lambda = 4/(3R) = 121.6 \text{ nm}$

Pattern 7: Particle in a Box

Find E_n and allowed wavelengths

$$E_n = n^2 h^2 / (8mL^2) \quad [\text{use } h = 6.626 \times 10^{-34}, m \text{ in kg, } L \text{ in m}]$$

$$\text{For electron in } L = 1 \text{ \AA: } E_1 = (6.626 \times 10^{-34})^2 / (8 \times 9.11 \times 10^{-31} \times (10^{-10})^2) = 6.03 \times 10^{-18} \text{ J} \approx 37.6 \text{ eV}$$

Pattern 8: Tunneling Probability

Find T through barrier

1. Find $\kappa = \sqrt{2m(V_0 - E)} / \hbar$
2. $T \approx e^{-2\kappa a}$ (a = barrier width)
3. Often asked: how does T change if a doubles? $\rightarrow T \rightarrow T^2$

SECTION E — PROFESSOR TRAPS (Conceptual Mistakes)

Unit I Traps

- ⚠ TRAP: Compton shift $\Delta\lambda = \lambda_C(1 - \cos\theta)$ is independent of incident wavelength λ . Never make $\Delta\lambda$ depend on λ .
- ⚠ TRAP: Phase velocity of matter wave is $v/2$, NOT v . Only group velocity equals particle velocity.
- ⚠ TRAP: Pair production requires a nucleus nearby. It CANNOT occur in empty space (momentum cannot be conserved with just photon + pair).
- ⚠ TRAP: 'Stopping potential' is the potential that stops the FASTEST (maximum energy) electrons. It's independent of intensity.
- ⚠ TRAP: The ultraviolet catastrophe is a classical failure at HIGH frequencies, NOT at low frequencies.
- ⚠ TRAP: De Broglie hypothesis applies to ALL matter, not just electrons. Bullets have a de Broglie wavelength too — just immeasurably small.

Unit II Traps

- ⚠ TRAP: Bohr energy $E_n = -13.6/n^2$ is NEGATIVE (bound state). Ionization energy is POSITIVE (+13.6 eV).
- ⚠ TRAP: Lyman series is in ULTRAVIOLET, not visible. Only Balmer ($n_1=2$) has visible lines.
- ⚠ TRAP: Rutherford's scattering formula gives probability proportional to $1/\sin^4(\theta/2)$, NOT $1/\sin(\theta)$.
- ⚠ TRAP: A 2-level laser is impossible: at best you achieve equal populations ($N_1 = N_2$), but never $N_2 > N_1$.
- ⚠ TRAP: The metastable state has a LONGER lifetime ($\sim 10^{-3}$ s) than ordinary excited states ($\sim 10^{-8}$ s). Students often reverse this.
- ⚠ TRAP: Rydberg constant $R = 1.097 \times 10^7 \text{ m}^{-1}$ (per meter). Remember to use consistent SI units.

Unit III Traps

- ⚠ TRAP: The operator acts on ψ first: $\langle p \rangle = \int \psi^*(\hat{p}\psi)dx$. Never apply $\hat{p}$ to ψ^* on the left.
- ⚠ TRAP: The TISE gives energy eigenvalues only for stationary states (time-independent V). Don't use it for time-varying problems.
- ⚠ TRAP: $|\psi|^2$ is real and non-negative. ψ itself can be complex. $|\psi|^2$ is NOT ψ^2 .
- ⚠ TRAP: Well-behaved wave function requires $d\psi/dx$ to be continuous EXCEPT at infinite potential boundaries.

Unit IV Traps

- ⚠ TRAP: PIB energy: $E_n = n^2 h^2 / (8mL^2)$. Use h (Planck's constant), NOT $\hbar$ (h -bar). Common error: using $\hbar$ gives a factor of $4\pi^2$ error.
- ⚠ TRAP: In a box, n starts at 1, NOT 0. $n=0$ gives $\psi=0$ (no particle). In harmonic oscillator, n starts at 0.
- ⚠ TRAP: Ground state energy of PIB is $h^2/8mL^2$, NOT zero. Zero-point energy is a consequence of confinement.
- ⚠ TRAP: Tunneling probability $T \approx e^{-2\kappa a}$ is approximate (WKB for square barrier). For variable $V(x)$, integrate $\kappa(x)$.

Unit V Traps

- ⚠ **TRAP:** Quantum teleportation does NOT violate no-cloning: Alice's original state is DESTROYED upon measurement.
- ⚠ **TRAP:** Teleportation requires 2 classical bits → cannot exceed speed of light. No FTL communication.
- ⚠ **TRAP:** Superdense coding: 1 qubit transmits 2 classical bits ($2 \rightarrow 1$ ratio, not $1 \rightarrow 2$). Students often confuse direction.
- ⚠ **TRAP:** Bell states cannot be written as product states. Any attempt to factor them gives a contradiction.
- ⚠ **TRAP:** Schmidt rank tells you entanglement: rank 1 = separable (not entangled); rank > 1 = entangled.
- ⚠ **TRAP:** CHSH: classical bound $S \leq 2$, quantum bound $S \leq 2\sqrt{2}$. Quantum $>$ Classical means quantum mechanics is 'more correlated.'

BEST OF LUCK IN YOUR EXAM!

You know this material. Trust your preparation.